

		Graph is a straight line, negative gradient but with a non-zero speed (as $r \neq 0$)
10	D	All geostationary satellites must have the same radius, and angular velocity $\omega = \frac{2\pi}{T}$, and hence they will have the same linear speed. Why C is incorrect $mr\omega^2 = \frac{GMm}{r^2} \quad \text{where } r - \text{radius of orbit}$ $T^2 = \frac{4\pi^2}{GM} r^3$ $(24 \times 60 \times 60)^2 = \frac{4\pi^2}{(6.67 \times 10^{-11})(6.0 \times 10^{24})} r^3$ $r = 4.231 \times 10^7 \text{ m}$ height from surface, $h = 4.231 \times 10^7 - R$
11	A	$GM(3M)$